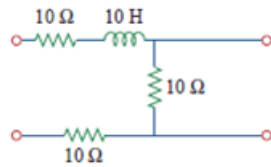


Problem

Obtain the z parameters for the network in Fig. 19.69 as functions of s .

Figure 19.69

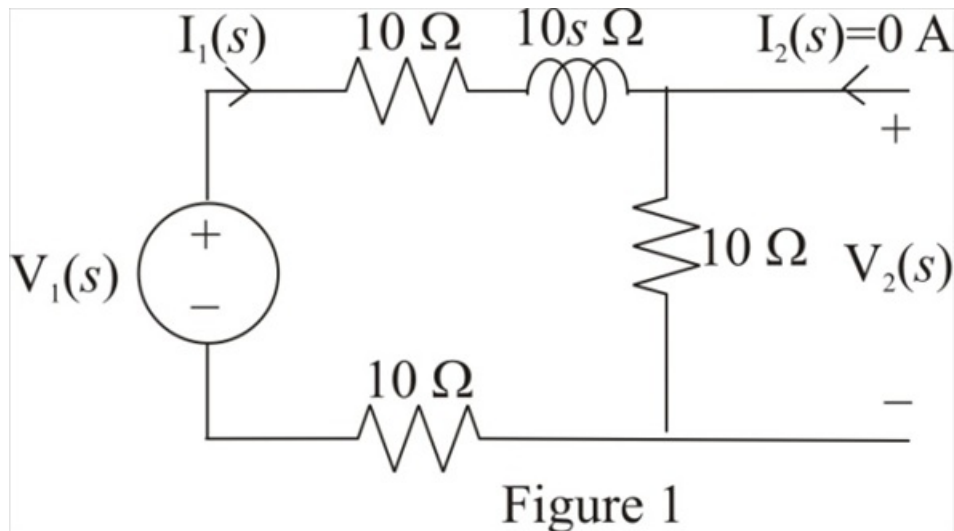


Step-by-step solution

Step 1 of 8

Refer to Figure 19.69 in the text book.

Connect the voltage source $V_1(s)$ as shown in Figure 1. Represent the parameters in Laplace notation. Identify the currents and voltages in the circuit.



Step 2 of 8

Consider the expression for the voltage $V_1(s)$.

$$\begin{aligned} V_1(s) &= I_1(s)(10 + 10s + 10 + 10) \\ &= I_1(s)(30 + 10s) \end{aligned}$$

Therefore, the expression for the voltage $V_1(s)$ is $I_1(s)(30 + 10s)$.

Consider the expression for the voltage $V_2(s)$.

$$V_2(s) = I_1(s)(10)$$

Therefore, the expression for the voltage $V_2(s)$ is $I_1(s)(10)$.

Step 3 of 8

Determine the expression for the open circuit input impedance $z_{11}(s)$.

$$z_{11}(s) = \frac{V_1(s)}{I_1(s)}$$

Substitute $I_1(s)(30+10s)$ for $V_1(s)$.

$$\begin{aligned} z_{11}(s) &= \frac{I_1(s)(30+10s)}{I_1(s)} \\ &= 30+10s \, \Omega \end{aligned}$$

Therefore, the expression for the open circuit input impedance $z_{11}(s)$ is $\boxed{30+10s \, \Omega}$.

Step 4 of 8

Determine the value of the open circuit transfer impedance $z_{21}(s)$ from port-2 to port-1.

$$z_{21}(s) = \frac{V_2(s)}{I_1(s)}$$

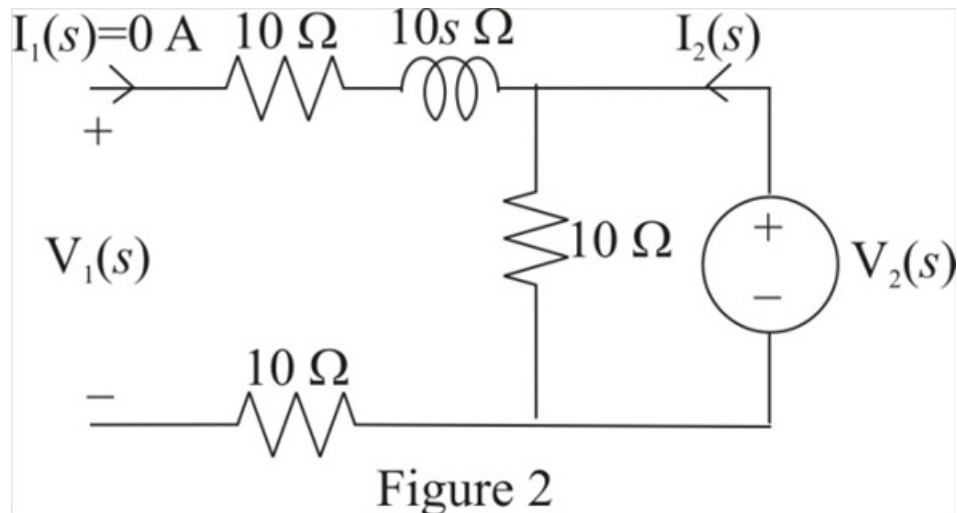
Substitute $I_1(s)(10)$ for $V_2(s)$.

$$\begin{aligned} z_{21}(s) &= \frac{I_1(s)(10)}{I_1(s)} \\ &= 10 \, \Omega \end{aligned}$$

Therefore, the value of the open circuit transfer impedance $z_{21}(s)$ from port-2 to port-1 is $\boxed{10 \, \Omega}$.

Step 5 of 8

Connect the voltage source $V_2(s)$ as shown in Figure 2. Identify the currents and voltages in the circuit.



Step 6 of 8

Consider the expression for the voltage $V_2(s)$.

$$V_2(s) = I_2(s)(10)$$

Therefore, the expression for the voltage $V_2(s)$ is $I_2(s)(10)$.

Consider the expression for the voltage $V_1(s)$.

$$V_1(s) = I_2(s)(10)$$

Therefore, the expression for the voltage $V_1(s)$ is $I_2(s)(10)$.

[Comments \(1\)](#)



Anonymous

is it not suppose to be $v_1(s) = I_1(s)(30 + 10s)$?

Step 7 of 8

Determine the value of the open circuit transfer impedance $z_{12}(s)$ from port-1 to port-2.

$$z_{12}(s) = \frac{V_1(s)}{I_2(s)}$$

Substitute $I_2(s)(10)$ for $V_1(s)$.

$$\begin{aligned} z_{12}(s) &= \frac{I_2(s)(10)}{I_2(s)} \\ &= 10 \, \Omega \end{aligned}$$

Therefore, the value of the open circuit transfer impedance $z_{12}(s)$ from port-1 to port-2 is

$$\boxed{10 \, \Omega}.$$

Step 8 of 8

Determine the expression for the open circuit output impedance $z_{22}(s)$.

$$z_{22}(s) = \frac{V_2(s)}{I_2(s)}$$

Substitute $I_2(s)(10)$ for $V_2(s)$.

$$\begin{aligned} z_{22}(s) &= \frac{I_2(s)(10)}{I_2(s)} \\ &= 10 \, \Omega \end{aligned}$$

Therefore, the expression for the open circuit output impedance $z_{22}(s)$ is $\boxed{10 \, \Omega}$.